Summary: redule. in[…] practice for scaling LLM: fraing

From scratch Re[…] [...]ing law: ~~smaller~~ model / more data

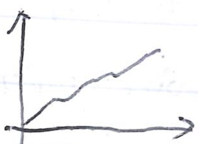
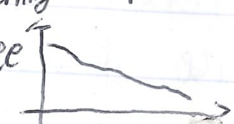
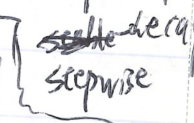
Distillat[…] [...]h to scaling actually work?

rewrite() into one

when fraing and fitting these things?

[...]ar Architectures / paramiterizations

[...]nal data-model ratio to scale nicely?

Cerebras - GPT	MiniCPM	Deepseek
⇒ 0.1 - 13B models	⇒ small, high-perf LM from Tsinghua group	⇒ 7B / 67B param models
⇒ trained with chinchilla recepi	⇒ 1.5B model, Beats most 2B/7B model	⇒ scale LLM with longerizm
⇒ more predicable scaling from mup params	⇒ careful, extensive. scaling computations + mup to stablize train	⇒ no any mup, directly estimate opimal batch / LR
⇒ mup parameerization combined with aggressive saling for hyperparametes	⇒ ~~opit~~ mup for initialicaom	 Batch size scaling curve
⇒ chachilla: small model + more data = good accuracy	fix aspect ratio, sale model	 learning rate scaling curve
⇒ scaling curve fits are good	⇒ optimal Batch size	⇒ dsk uses chichalla for fraing LLM
	⇒ bach size ↑ loss size ↓	WSD-style ↓ (warm / ~~stable~~ ~~stable~~ decay) stepwise
	⇒ optimal / stable LR	⇒ WSD. warm up stepwise decay
	⇒ WSD learning rate	
	split LR into 3 phases: warmup, stable, decay	⇒ Dsk: ~~rela~~ replaces the typical cosine learning rate decay with stepwise-decay learning rate
	Llama ⇒ optimal model-data ratio	 WSD chinchilla

Development:

Kaplan's scaling law ⟶ chanchilla's optimal scaling law

↓ performance sale with size / compute

↓ data / param ratio ⟶ MuP stablize LR / gradient